

for (i=0; i < n; i++) — $O(n)$

for (i=0; i < n; i=i+2) — $\boxed{\frac{n}{2}} \Rightarrow O(n)$

for (i=n; i > 1; i--) — $O(n)$

for (i=1; i < n; $\boxed{i=i*2}$) — $O(\log_2 n)$

for (i=1; i < n; $\boxed{i=i*3}$) — $O(\log_3 n)$

for (i=n; i > 1; i=i/2) — $O(\log_2 n)$

void fact1(int n, list<Integer> perm)

{ if (!used[j])

{ fact1(n, perm) → n $O(n!)$

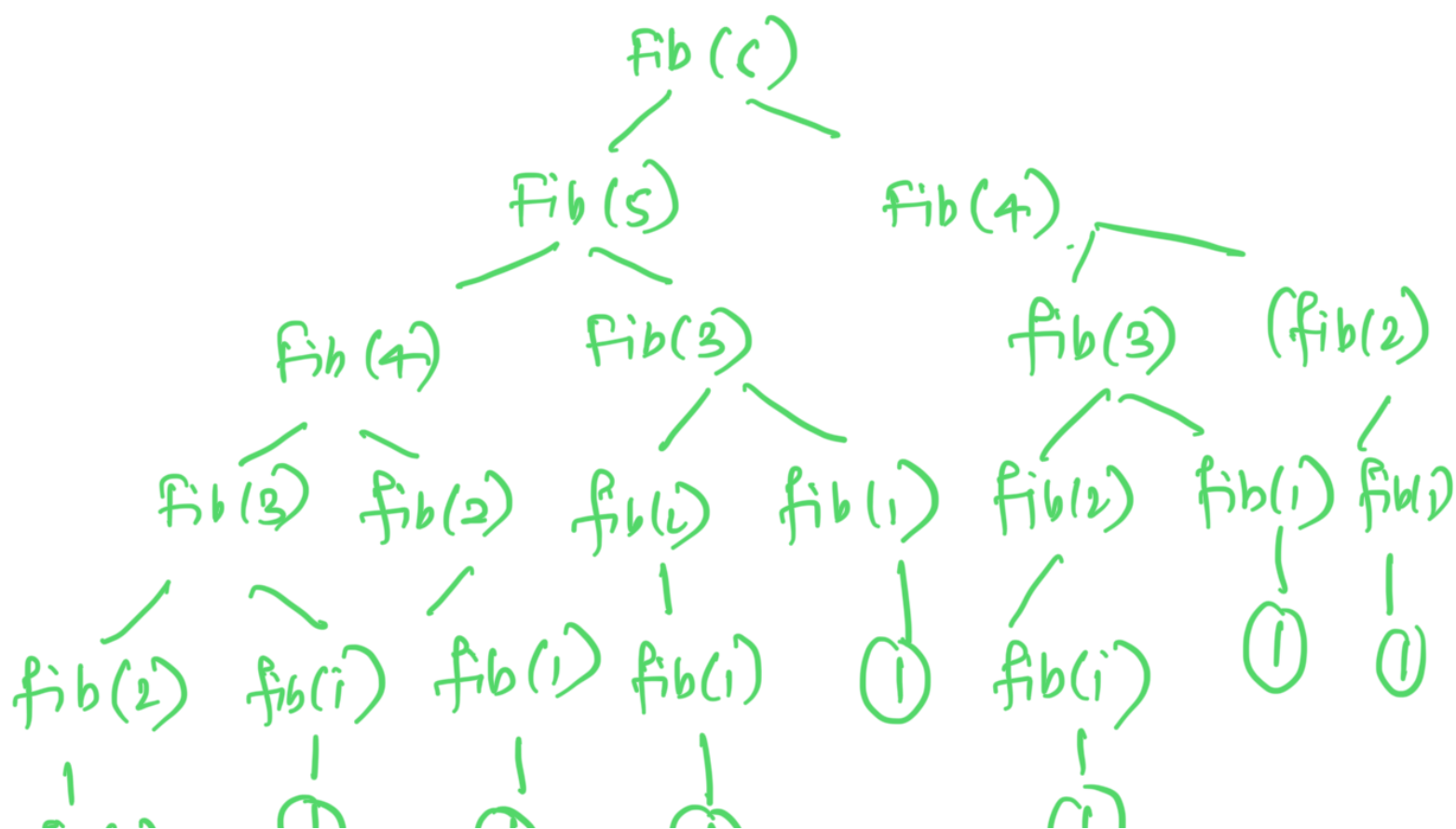
void fib(int n)

{ if (n <= 1)

return n;

return fib(n-1) + fib(n-2); }

Exponential time complexity → $\boxed{O(2^n)}$



$\text{fib}(1)$ $\text{fib}(2)$ $\text{fib}(3)$ $\text{fib}(4)$ $\text{fib}(5)$
 ①

Types of Time Complexities

$O(1) \rightarrow$ Constant

$O(\log n) \rightarrow$ Logarithmic

$O(n) \rightarrow$ Linear

$O(n^2) \rightarrow$ Quadratic

$O(n^3) \rightarrow$ Cubic

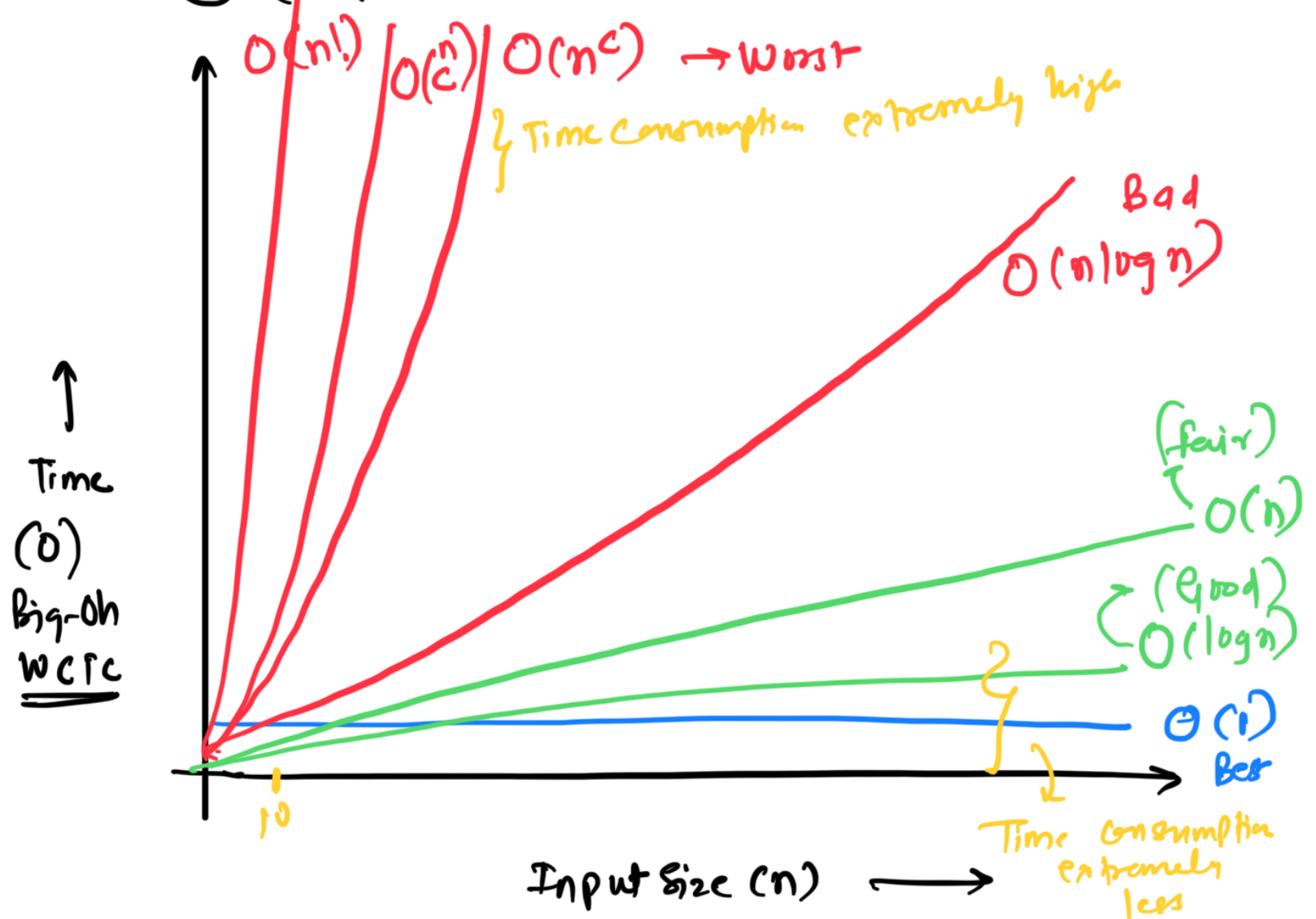
$O(2^n) \rightarrow$ Exponential

$O(3^n) \rightarrow$ Exponential

$O(n^n) \rightarrow$ Exponential

$O(\sqrt{n}) \rightarrow$ Square root

$O(n!) \rightarrow$ Factorial



$$\boxed{O(1)} < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < \dots$$

Best $2^n < 3^n < \dots < \boxed{n^n}$ Worst

String Reverse $\rightarrow O(n^2)$ - 2 for loop - ① ✓
 $\rightarrow O(n)$ - 2 ptr - ② ✓
 $\rightarrow \underline{O(1)}$ - Collection (optimized) - ③ ✓

Best case, Worst case, Average

① Linear Search -

A $\rightarrow$

8	6	12	5	9	7	4	13	16
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0 1 2 3 4 5 6 7 8

Key $\rightarrow$ 8

①st Comp $\Rightarrow$ Match
 Best case $\rightarrow O(1)$ ① $\Rightarrow$ Ans. Min

Key $\rightarrow$ 4

Comp $\Rightarrow$ 7

No of elements $= \frac{9}{2} = 4.5 \approx 5$

Average = 5 Comparison

$$A(n) \Rightarrow \frac{n+1}{2} = O(n)$$

5 (> 7) $\approx$ Average.

$\leftarrow$ Average case $\rightarrow O(n)$

Key $\rightarrow$ 16

Comp $\Rightarrow$ 9 $\leftarrow$ Max
 No of elem $\Rightarrow$ 9

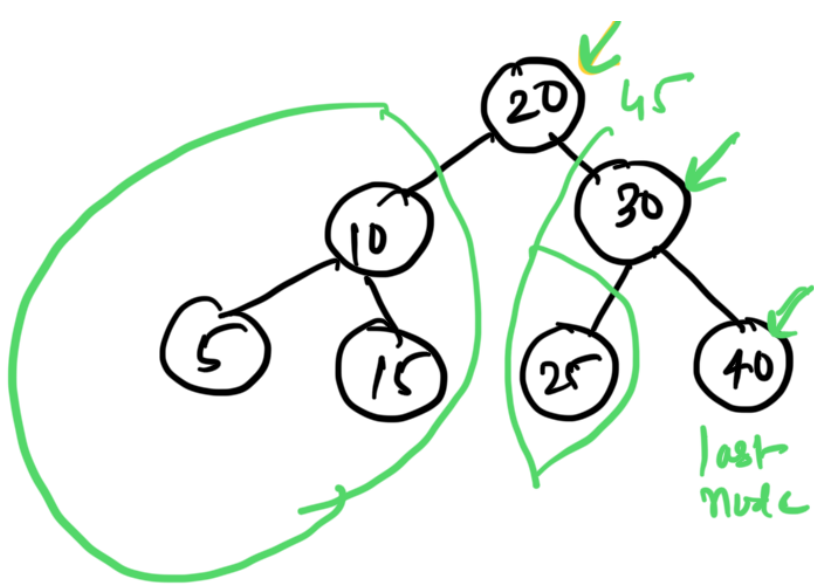
Worst case $\rightarrow O(n)$

Key $\rightarrow$ 25

Comp $\Rightarrow$ 9 $\leftarrow$ Max comp.
 No of elem $\Rightarrow$ 9

Worst case $\rightarrow O(n)$

Ex: Binary Search Tree



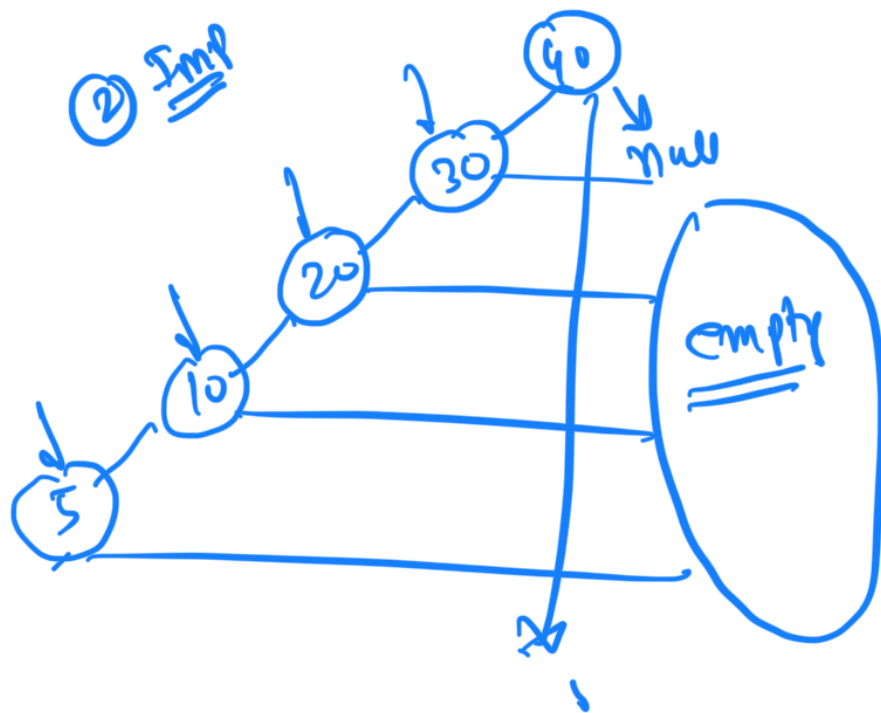
Key = 20

No. Comp = 1

Best case $\Rightarrow O(1)$

Key = 45 $\rightarrow$ Not found

No of = $n/2$



Worst ($\log n$)
_Case

Key = 5 Worst
 $O(n)$ Case

$\uparrow$
Max no comp